

SQL foundations

Exercise 1

employees						
id	first_name	last_name	department	salary	hire-date	city
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

1. Retrieve all columns from the employees table.

```
SELECT *  
FROM employees;
```

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3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

2. Find all unique departments

```
SELECT DISTINCT department  
FROM employees;
```

department
IT
HR
Finance
Marketing

3. Retrieve first and last names ordered by salary descending

```
SELECT first_name,  
       last_name  
FROM employees  
ORDER BY salary DESC;
```

first_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	McKoen
Brian	Lee

4. Retrieve the top 3 highest-paid employees.

```
SELECT id,  
       first_name,  
       last_name,  
       salary  
FROM employees  
ORDER BY salary DESC  
LIMIT 3;
```

id	first_name	last_name	salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

5. Find employees in the IT department

```
SELECT id,  
       first_name,  
       last_name,  
       department
```

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

```
FROM employees  
WHERE department = 'IT';
```

6. Find employees in finance with salary > 60000.

```
SELECT id,  
       first_name,  
       last_name,  
       department,  
       salary
```

```
FROM employees
```

```
WHERE department = 'finance' AND salary > 60000
```

id	first_name	last_name	department	salary
3	Cathy	Zulu	finance	65000

7. Find employees in HR or Marketing

```
SELECT id,  
       first_name,  
       last_name,  
       department
```

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

```
FROM employees
```

```
WHERE department = 'HR' OR department = 'Marketing';
```

8. Find employees not in IT

```
SELECT id,  
       first_name,  
       last_name,  
       department
```

```
FROM employees
```

```
WHERE department != 'IT';
```

id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

9. Find employees in HR, IT, Or finance using IN

```
SELECT id,  
       first_name,  
       last_name,  
       department  
FROM employees  
WHERE department IN ('HR', 'IT', 'finance');
```

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	finance
5	Eva	Naidoo	IT

10. Find employees in IT with salary > 65000 and city Johannesburg

```
SELECT id,  
       first_name,  
       last_name,  
       department,  
       salary,  
       city  
FROM employees  
WHERE department = 'IT' AND salary > 65000 AND city = 'Johannesburg';
```

id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg